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Forgetting-aware presentation of GenAI summaries to support task resumption in cross-session academic search

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Abstract

When an academic search task spans multiple sessions, generative AI embedded in the search interface can help the searcher to consolidate what they have found via a summary that supports *task resumption*. Such summaries are typically based on the resources the searcher saved. However, a searcher's memory decays unevenly across the gap between sessions, with shallowly engaged material most likely to be forgotten. We propose *Ebb*, a lightweight mechanism that adapts the presentation of the task resumption summary to the searcher's predicted memory state. Material the searcher engaged with deeply is compressed to a brief reminder; material they skimmed, or accepted without checking, is expanded, with each claim linked to the saved source that supports it. The prediction uses a heuristic over interaction signals logged by the search interface. The primary contributions are the search interface design and heuristic details.

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Keywords: cross-session search; task resumption; generative AI; academic search; search as learning; cognitive offloading

1. Introduction

Academic search tasks that involve learning are rarely completed in one sitting: searchers gather material, pause for days, and must then re-enter a partially built mental model of their search [1]. Near the end of a session, searchers often undertake a *consolidation* activity: they distill what they have saved into notes or summaries, and this consolidated material is the first thing they see when the task is resumed. Generative AI (genAI) is well suited to supporting this consolidation activity, but its efficiency has a cognitive price: the more the system does the consolidating, the less deeply the searcher processes the material [2]. A recent controlled study of cross-session academic search [3], set in an aggregated search interface with a multi-workspace environment that supports human-controlled clustering [4], compared two consolidation *scaffolds*, that is, interface supports that structure the consolidation work: steerable genAI summaries and human-authored notes. It found contrasting profiles: genAI summaries brought faster resumption, broader conceptual coverage, and higher self-reported knowledge; self-authored notes brought richer inter-concept connections and higher certainty. Some participants also accepted genAI claims without checking their sources.

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Both scaffolds in that study share one property: the artifact is fixed *across the gap between sessions*. Interfaces that vary the effort they demand do exist, but they act within a single session on material in front of the reader [5]; none adapts its presentation at the time of task resumption in response to what happened during the search and the length of the interval between the search sessions. Yet across that gap the searcher's memory decays unevenly, and two robust regularities predict where: retention declines with the time between the activities [6], and shallowly engaged material is the most likely to have become irretrievable, its weaker encoding leaving less retrieval strength to lose [7]. A static artifact therefore misallocates support at the moment it matters most: it expends screen space on material the searcher still knows while under-supporting material they have forgotten.

We propose *Ebb*, which adapts the presentation of search activity summaries to the searcher's predicted memory state: deeply engaged material is compressed to brief reminders, while shallowly engaged material is expanded, with each claim linked to its saved source and with unverified claims flagged.

2. Background and motivation

Cross-session search and the cost of coming back. Multi-session tasks are frequent and expected in many search contexts [1, 8], and re-entry is a recognized bottleneck [9]. Cognitive accounts model task resumption as the decay of goal activation over the suspension interval, with speed depending on the quality of available retrieval cues [10].

GenAI consolidation and cognitive offloading. When GenAI writes the consolidation, part of the learner's cognitive work is offloaded with it [11]: people judge their competence poorly during low-effort study [12], and over-rely on AI outputs unless the interface prompts deliberation [13]. Our motivating study found that how consolidation labour is split between learner and system shapes which aspects of learning are supported [3].

Forgetting models and scheduled reminders. Quantitative models of forgetting have a long history [6] and underpin mature technologies that follow two general approaches. Spaced repetition schedules item reviews against per-learner forgetting curves [14]. Adaptive hypermedia varies presentation against a modelled knowledge state [15].

3. Ebb: Forgetting-aware presentation of consolidated material

None of the technologies above adapt the searcher's own consolidated material, deciding at task resumption time how much of it to show and at what level of granularity, with support allocated inversely to predicted retention. Ebb fills this gap. This work is built upon a foundation of a workspace environment that includes clusters of saved resources [3]. While current approaches to presenting consolidation artifacts focus on what was found, Ebb focuses on what the learner is predicted to retain in their long-term memory: thin information where memory is strong, dense information where it has ebbed. In adaptive-presentation terms [15], the user model is a forgetting model.

The forgetting heuristic. For each cluster of search results c within a workspace, Ebb computes a risk score $R_c = (1 - E_c)\, g(\Delta t)$. The engagement aggregate $E_c = \sum_i w_i \min(s_i^{(c)}/\kappa_i, 1)$ is a weighted sum of the signals in Table 1, each normalized by a saturation cap κ_i (for example, four minutes of dwell counts as full engagement); with weights summing to one, $E_c \in [0, 1]$ exactly. The delay factor $g(\Delta t) = 1 - e^{-\Delta t/\tau}$ grows with the time since the previous session ended; $\tau = 3$ days places the steep segment of the decay inside the first week, mirroring the forgetting curve's early losses [6], and is a candidate for calibration against observed forgetting.

Absolute thresholds ($R_c < 0.25$ compress; $R_c \geq 0.55$ expand; keep otherwise) anchor the presentation policies to predicted memory state rather than to a within-learner ranking, so a searcher who processed everything deeply sees everything compressed.Because every cluster shares the same gap at a given return, differences between clusters come from engagement, while the gap governs how strongly the whole artifact adapts: after a ten-minute break, no cluster comes near the expand threshold. Weights are fixed in advance (dwell 0.35, edits 0.25, verification 0.25, revisits 0.10, authored text 0.05).

This approach represents a first-order proxy, not a cognitive model of the learner, motivated by two regularities: deeper engagement predicts more durable encoding [7], and retention declines with time [6].

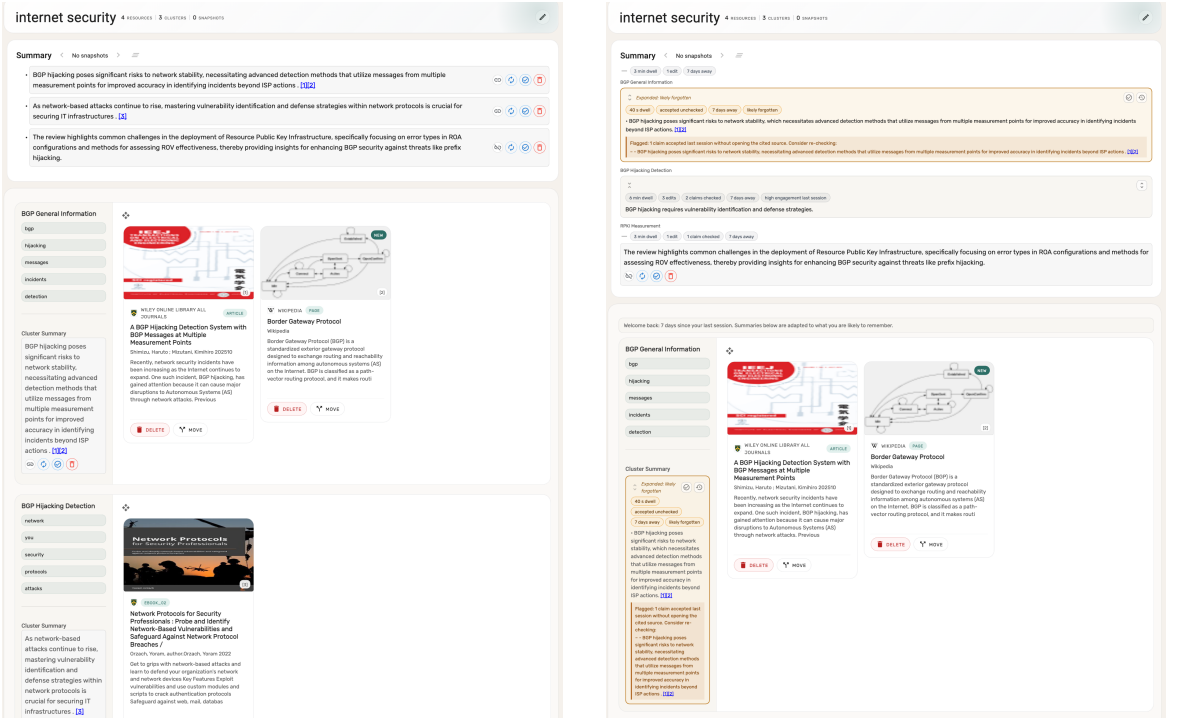
Interaction design. Figure 1 shows Ebb in the interface of our motivating study [3], whose workspace organizes saved results into topical clusters; the mechanism itself requires only a sectioned summary and logged interaction, so it is not tied to aggregated search or any specific underlying search engine. During a search session, the workspace logs the signals documented in Table 1. At the time of task resumption, each cluster is presented according to its risk

Table 1: Per-cluster engagement signals feeding E_c. Elapsed time enters the risk score only through the delay factor $g(\Delta t)$. The first four signals are already logged by a typical search interface that includes a workspace; verification events require claims to be linked to their sources, the same linkage that enables the unverified-claim flag.

Signal	Example	Interpretation
Dwell time on cluster content	6 min vs. 40 s	Depth of processing [7] (proxy; may reflect difficulty)
Edits and corrections to summary text	3 inline corrections	Generative engagement [16]
Learner-authored text	a written annotation	Self-generated encoding [16]
Revisits within the session	cluster reopened twice	Spaced re-exposure [17]
Verification events	source opened from a claim	Critical engagement [13]

score. Low-risk clusters *compress*: the cluster's title stays exactly as the searcher left it, serving as a stable memory anchor, and the genAI summary shrinks to a one-line reminder, restorable with one click. The compressed state is labelled with the recorded activity behind the decision (time spent, edits made, claims checked) rather than with any claim of mastery, which could inflate the searcher's confidence [12]. Moderate-risk clusters are *kept* as-is. High-risk clusters *expand*: fuller phrasing, each claim linked to the saved source that supports it, and a flag on claims the searcher accepted without opening their source in the previous session.

Guided by design principles for exploratory search interfaces [18], every presentation decision is marked with a compact state indicator and explained by the signals that caused it. In addition, the expanded summaries are highlighted in orange to draw attention to them. Every decision is reversible, including single-click controls that mark clusters as remembered and inform later sessions. The mechanism is also lightweight: one generation call at resumption over per-cluster metadata local to the workspace, an incremental layer on existing genAI summary scaffolds. As that call rewrites text and could add unsupported phrasing, expanded presentations keep source anchors verbatim and remain fact-checkable like the original.



(a) Workspace view at the end of the search session.

(b) Workspace view upon resuming the search one week later.

Fig. 1: Screenshots of how Ebb's workspace changes from initial search (a) to resumed search (b).

Figure 1b showcases the three states of genAI summarization. The first is expanded because, in the initial search session, the searcher spent less than one minute evaluating the saved resources, including having accepted a claim without opening the cited source. The assessment is that the information was likely forgotten. The second is compressed due to a high degree of engagement with the information in the initial session. The third is left unchanged due to insufficient evidence to support compression or expansion.

4. Conclusion

Ebb represents a reconsideration of how genAI can be used to generate summaries to support task resumption. It allows for efficiency where memory is strong and engagement where it is weak, instead of choosing between the two. It treats the searcher's memory state as an explained, reversible input to how the summaries are presented, and makes the artifact an inspectable judgment of what the searcher is expected to no longer know, contestable with a click.

What remains to be studied is whether searchers find value in how Ebb personalizes the summaries at task resumption time. This will be assessed in forthcoming laboratory and longitudinal user studies. Forgetting-aware presentation could offer a general design resource for cross-session search: genAI support that knows not only what was found, but what should be resurfaced to support task resumption.

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